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Title of Invention

LIQUID CONTAINER ASSEMBLY WITH REMOVABLE OPTICAL PROJECTION
BASE, AND PROJECTION BASE MODULE

Technical Field

[0001] The present disclosure relates to a liquid container assembly and consumer electronic devices, and more particularly to a liquid container assembly incorporating a removable electronic projection module (projection base module) configured to generate, by means of an optical member, a visual effect in which an image is perceived to float within or above the container (hereinafter referred to as a “floating-image effect” or “pseudo-holographic effect”), and to the projection base module.

Background Art

[0002] Conventional drinking containers are passive articles that provide little user interaction beyond containing a beverage. In contrast, beverage containers integrated with display devices and various electronic components have recently been proposed (see, for example, Patent Documents 1 and 2). For example, Patent Document 1 discloses a transparent liquid container in which a fine uneven pattern is formed on a side surface thereof, and a hologram image is made visible by utilizing reference light from a coaster or the like and changes in the refractive index of a liquid. However, the configuration disclosed in Patent Document 1 is merely an effect relying on static interference fringes engraved on the glass surface, and cannot display dynamically changing image content. In addition, a configuration that requires a separate coaster or the like emitting reference light, as in Patent Document 1, tends to make portability and handling cumbersome.

[0003] Patent Document 2 discloses an entertainment beverage glass that presents a virtual image by displaying an image from an image display device fixed to a side surface of a glass main body through liquid and a curved portion. However, although the configuration disclosed in Patent Document 2 shows a virtual image using a lens effect, it has an external structure in which the image display device is fixed to a flat

portion on the side surface via an adjuster mechanism, a pocket mechanism, or the like. Therefore, the structure is complex, and there is room for improvement in terms of handling and aesthetic design.

Citation List

[0004]

[Patent Document 1] Japanese Unexamined Patent Application Publication No. H08-70974

[Patent Document 2] Japanese Patent No. 6679076

Summary of Invention

Technical Problem

[0005] An object of the present disclosure is to provide a liquid container assembly and a projection base module used therefore capable of generating a floating-image visual effect observable within the container or above the container, despite having a simple structure, while maintaining waterproof separation of the electronic components from the liquid-contacting parts, ease of cleaning, modularity permitting the electronic module to be detached and reused, and wireless operation including wireless charging and wireless content updating. Note that in the present disclosure, the term "liquid container" (or "liquid container assembly") refers to containers in general having a structure capable of containing or holding a liquid therein, and includes, but is not limited to, beverage containers such as glasses and cups, vases, ornamental containers, and aroma diffuser containers.

Solution to Problem

[0006] According to one aspect of the present disclosure, there is provided a liquid container assembly comprising: a container body configured to contain a liquid; a base assembly removably coupled to the container body; a sealing member disposed between the container body and the base assembly and configured to inhibit ingress of liquid into the base assembly; a display module housed in the base assembly and configured

to emit image light; an optical member arranged to redirect the image light such that a virtual image is perceived to float within or above an interior region of the container body; a rechargeable battery and a wireless charging receiver housed in the base assembly; and wireless communication circuitry and a controller housed in the base assembly, the controller being configured to receive image content from an external device and to cause the display module to display the image content. All electrically powered components of the assembly are housed in the base assembly, such that the container body is free of electrically powered components. It is the integration of these elements into a single removable, sealed, wirelessly charged base module wirelessly updatable with respect to image content and cooperating with an electronics-free container body that distinguishes the present disclosure from known illuminated drinkware.

[0007] According to another aspect, there is provided a projection base module, per se, removably attachable to a container body, the module housing the display module, a rechargeable battery, a wireless charging receiver, wireless communication circuitry, a controller, and a memory storing image content.

Advantageous Effects of Invention

[0008] Because the electronic components are housed entirely within the removable base assembly and are separated from the container body by the sealing member, the container body can be washed independently of the electronics, the battery can be recharged or the module serviced without exposing electronics to liquid, and a single base module can be used with a plurality of container bodies. The optical member enables a floating-image effect not achievable with simple illumination, and the wireless communication circuitry permits the image content to be updated from an external device without any physical connector penetrating the waterproof boundary.

Brief Description of Drawings

[0009] FIG. 1 is a perspective view of a liquid container assembly according to an embodiment of the present disclosure.

[0010] FIG. 2 is an exploded view illustrating the container body, the sealing member,

and the removable base assembly of FIG. 1.

[0011] FIG. 3 is a perspective view illustrating an observation mode of a virtual image (floating image) generated by an optical member in a liquid container assembly according to an embodiment of the present disclosure.

[0012] FIG. 4 is a block diagram illustrating a configuration of an electronic system of a projection base module (base assembly) according to an embodiment of the present disclosure.

[0013] FIG. 5 is a vertical cross-sectional view illustrating a liquid container assembly according to an embodiment of the present disclosure.

Description of Embodiments

[0014] Hereinafter, embodiments of the present disclosure will be described with reference to the drawings. The embodiments described below are illustrative, and the present disclosure is not limited thereto.

[0015] Referring to FIGS. 1 and 2, a liquid container assembly 100 comprises a container body 1 configured to contain a liquid and a base assembly 2 removably coupled to a bottom portion of the container body 1. The container body 1 is preferably formed, at least in part, of a transparent or translucent material such as glass, borosilicate glass, Tritan, polycarbonate, or acrylic resin, so that an interior region of the container body 1 provides an optical cavity through which the floating image can be observed. The container body 1 may take any of a variety of form factors, including without limitation a cup, a tumbler, a wine glass, a cocktail glass, a mug, a stein, a bottle, a carafe, a decanter, or a pitcher; the present disclosure is not limited to any particular container body shape, height, or volume, and the base assembly 2 may be dimensioned to suit any such form factor.

[0016] A bottom portion of the container body 1 includes an engagement structure 3 for detachably coupling with the base assembly 2. In one embodiment, the engagement structure 3 is an internal thread (female thread) formed on an inner peripheral surface of a lower bottom margin of the container body 1, and is configured to be threadedly

engaged with a complementary structure 4 (for example, a complementary external thread (male thread)) formed on an outer peripheral surface of the base assembly 2. In other embodiments, the engagement structure 3 may comprise a bayonet coupling, a magnetic coupling, or a snap-fit coupling, and any releasable mechanical coupling that consistently determines relative positions in a circumferential direction (rotational direction) and an axial direction between the container body 1 and the base assembly 2 may be employed. By having a circumferential positioning structure, such as a polygonal shape, a D-cut shape, or a keyed shape with inter-fitting projections and recesses, the engagement structure 3 (and the complementary structure 4) can repeatably (consistently) maintain a rotational position in the circumferential direction of the container body 1 relative to an optical member 6 described below, so that a floating image is presented at a constantly intended fixed position relative to the container body 1.

[0017] The base assembly 2 houses the electronic components required to generate the floating image. Disposed within the base assembly 2 is a display module 5 configured to emit image light. The display module 5 may comprise a liquid crystal display (LCD), an organic light-emitting diode (OLED) display, a micro-LED display, a micro-OLED display, or any equivalent light-emitting display device. The display module 5 may further include a touch controller (touch control unit) for changing or controlling at least one of presence/absence of display (ON/OFF), presence/absence of animation display (ON/OFF), a color tone of emitted light, and brightness of light.

[0018] In one embodiment, a protective layer (a protective sheet, a protective film, or a protective plate such as an acrylic plate) for protecting the display module 5 from penetration of humidity, moisture, or liquid may be laminated or disposed on an emission surface side of the display module 5. Thereby, waterproof and moisture-resistant performance against moisture resulting from condensation on the container body 1, unexpected liquid leakage, or the like can be further enhanced.

[0019] An optical member 6 is positioned above the display module 5 (or the protective layer) (preferably directly above an emission surface thereof) in an optical path of the image light. In one embodiment, the optical member 6 comprises a prism. In other embodiments, the optical member 6 may comprise a partially reflective plate, a beam

splitter, an inclined half-mirror, a pyramidal reflector, or a combination thereof. The optical member 6 is configured to receive the image light emitted from the display module 5 (for example, vertically upward) and deflect (reflect or refract) the image light by its own refractive surface or reflective surface (an inclined surface or the like), thereby forming a virtual image (a pseudo-holographic image) of the image content in an internal region of the container body 1 or in a spatial region above it. Thereby, an observer can visually observe (perceive) the virtual image of the image content as if floating inside the container body 1 or in a region above it. Such perception can be obtained from a wide range of viewing angles, such as observation from the entire periphery (360 degrees) through a transparent or translucent wall of the container body 1, observation through an observation window formed in an opaque wall, or observation from above through an open top of the container body 1. Thereby, the observer can visually observe a three-dimensional floating image from any position in the surroundings or from a line of sight looking down. It is noted that the resulting effect is a pseudo-holographic (virtual-image) effect based on reflection or refraction of the image light, rather than a hologram in the strict interferometric sense; the terms “floating image” and “pseudo-holographic” are used herein in this meaning.

[0020] A sealing member 7 is disposed between the container body 1 and the base assembly 2. The sealing member 7 is preferably a resilient elastomeric member, for example an O-ring formed of silicone rubber or a thermoplastic elastomer, and is compressed between both when the container body 1 and the base assembly 2 are coupled, thereby inhibiting ingress of liquid, moisture, or washing water into an electronic compartment of the base assembly 2. In one embodiment, the sealing member 7 may be disposed so as to be radially compressed between an inner peripheral portion of the engagement structure 3 and an outer peripheral portion of the base assembly 2, or between an outer peripheral portion of the engagement structure 3 and an inner peripheral portion of the base assembly 2. In another embodiment, the sealing member 7 may be disposed so as to be axially compressed between a lower surface of the engagement structure 3 (or a bottom portion of the container body 1) and an upper surface (or a receiving step) of the base assembly 2, or an arrangement combining these may be employed. In one embodiment, the base assembly 2 with the display module 5

sealed therein attains an ingress protection rating of at least IPX5.

[0021] The base assembly 2 further houses: a rechargeable battery 8; a wireless charging receiver 9; wireless communication circuitry 10; a controller 11; and a memory 12. The rechargeable battery 8 is preferably a lithium-ion or lithium-polymer battery. The wireless charging receiver 9 is, for example, a receiver conforming to the Qi standard, and may include a charge control circuit (charge controller) and a wireless charging coil. Thereby, based on power received via the wireless charging coil, the rechargeable battery 8 can be charged via the charge control circuit without providing any external connector penetrating the housing of the base assembly 2. The wireless communication circuitry 10 supports at least one of Bluetooth, Bluetooth Low Energy, Wi-Fi, and near-field communication (NFC). The controller 11 is, for example, a microcontroller. The memory 12 is, for example, a flash memory, and stores image content to be displayed by the display module 5.

[0022] The controller 11 is configured to read image content from the memory 12 and to drive the display module 5 to display the image content. The controller 11 is further configured to receive, via the wireless communication circuitry 10, image content or control data transmitted from an external device 200 such as a smartphone, tablet, or personal computer, and to store the received image content in the memory 12 and/or update the content displayed by the display module 5 in response thereto. The image content may include static images, animations, logotypes, text, rendered three-dimensional objects, promotional content, or user-generated graphics. The controller 11 may include a setting button (an operation unit such as a switch or a touch panel) for changing at least one setting among an operation mode, a production pattern, a timer, and volume. Furthermore, the controller 11 may include a speaker that reproduces audio such as music, sound effects, environmental sounds, or voice messages in conjunction with display of the image content or changing of the settings.

[0023] In a further embodiment, the base assembly 2 may further include a thermoelectric conversion element, such as a Seebeck element, configured to generate electrical power utilizing a temperature difference between the liquid contained in the container body 1 and the interior of the base assembly 2 (or ambient air). The thermoelectric conversion element may be thermally coupled to, for example, a bottom

wall of the container body 1 and configured to generate electricity based on a temperature difference between the liquid and ambient air, thereby providing supplemental power to recharge the rechargeable battery 8 or to drive electronic components such as the display module 5.

[0024] In a further embodiment, the controller 11 changes the displayed image content in response to a detected condition, for example a signal from a sensor housed in the base assembly 2 (such as an accelerometer, a temperature sensor thermally coupled to the bottom of the container body 1, or a capacitive liquid-level sensor), so that the floating image responds to lifting of the assembly, to the temperature of the liquid, or to the liquid level.

[0025] In a further embodiment, the container body 1 is of double-walled construction, the floating image being formed within a cavity defined between the inner and outer walls or within the liquid-containing interior, and the bottom wall of the container body 1 that faces the optical member 6 is optically transmissive.

[0026] A characterizing feature of the present disclosure is that all electrically powered components of the liquid container assembly 100 — including the display module 5, the battery 8, the wireless charging receiver 9, the wireless communication circuitry 10, the controller 11, and the memory 12 — are housed within the base assembly 2, such that the container body 1 is entirely free of electrically powered components. Because the base assembly 2 is removable, the container body 1 — which alone contacts the liquid — can be washed by hand or in a dishwasher without exposing any electronic component to water, and the base assembly 2 can be attached to any container body having a complementary coupling, permitting a single electronic module to serve a family of differently shaped container bodies of any of the form factors mentioned above.

[0027] Note that the base assembly 2 in the present disclosure can be configured as a "projection base module" detachably attachable to the above-described container body 1. This projection base module (base assembly 2) includes: a housing having a coupling portion (for example, a threaded portion or the like) detachably engaging a bottom portion (engagement structure 3) of the container body 1; and the display module 5, the optical member 6, the rechargeable battery 8, the wireless charging receiver 9, the

wireless communication circuitry 10, the memory 12, and the controller 11 housed in the housing. In addition, the projection base module may further include a sealing member 7 (for example, an O-ring) that is compressed between the housing and the container body 1 when coupled to the container body 1, thereby inhibiting (sealing) ingress of liquid into the housing. The optical member 6 is arranged so as to, when the projection base module is coupled to the container body 1, deflect image light from the display module 5 and cause a virtual image to be perceived as floating within or above an internal region of the container body 1. Further, the controller 11 causes the display module 5 to display image content received from an external device via the wireless communication circuitry 10.

Industrial Applicability

[0028] The disclosed liquid container assembly and projection base module are applicable to consumer drinkware, hospitality and bar equipment, promotional and advertising merchandise, and event and entertainment products, and can be manufactured using existing display, molding, and sealing technologies.

Reference Signs List

[0029]

100 Liquid container assembly

1 Container body

2 Base assembly (projection base module)

3 Engagement structure

4 Complementary structure (complementary engagement portion)

5 Display module

6 Optical member

7 Sealing member

8 Rechargeable battery

9 Wireless charging receiver

10 Wireless communication circuitry

11 Controller

12 Memory

200 External device

【書類名】 外国語特許請求の範囲

[Claim 1] A liquid container assembly comprising:

a container body configured to contain a liquid;

a base assembly removably coupled to the container body;

a sealing member disposed between the container body and the base assembly and configured to inhibit ingress of liquid into the base assembly;

a display module housed in the base assembly and configured to emit image light;

an optical member positioned above the display module and configured to deflect the image light emitted from the display module to form a virtual image within or above an interior region of the container body;

a rechargeable battery and a wireless charging receiver housed in the base assembly, the wireless charging receiver being configured to charge the rechargeable battery without a wired connection penetrating the base assembly; and

wireless communication circuitry and a controller housed in the base assembly, the controller being configured to receive image content from an external device via the wireless communication circuitry and to cause the display module to display the image content,

wherein all electrically powered components of the container assembly are housed in the base assembly, such that the container body is free of electrically powered components.

[Claim 2] The liquid container assembly according to claim 1, wherein the optical member comprises at least one of a prism, a partially reflective plate, a beam splitter, an inclined half-mirror, and a pyramidal reflector, or a combination thereof.

[Claim 3] The liquid container assembly according to claim 1, wherein the display module comprises a liquid crystal display, an organic light-emitting diode display, a micro-LED display, or a micro-OLED display.

[Claim 4] The liquid container assembly according to claim 1, wherein the base assembly is removably coupled to a bottom portion of the container body via an

engagement structure formed at the bottom portion of the container body.

[Claim 5] The liquid container assembly according to claim 4, wherein the engagement structure comprises a threaded coupling, a bayonet coupling, a magnetic coupling, or a snap-fit coupling.

[Claim 6] The liquid container assembly according to claim 4, wherein the engagement structure includes a circumferential positioning structure that defines an orientation of the container body relative to the optical member.

[Claim 7] The liquid container assembly according to claim 1, wherein the sealing member is an elastomeric O-ring compressed between the container body and the base assembly when the container body and the base assembly are coupled.

[Claim 8] The liquid container assembly according to claim 1, wherein the base assembly further houses a memory, and the controller is configured to store the image content received from the external device in the memory.

[Claim 9] The liquid container assembly according to claim 1, wherein at least a bottom wall of the container body facing the optical member is optically transmissive, and at least a portion of a side wall of the container body is transparent or translucent.

[Claim 10] The liquid container assembly according to claim 1, wherein the container body is a cup, a tumbler, a wine glass, a cocktail glass, a mug, a stein, a bottle, a carafe, a decanter, or a pitcher.

[Claim 11] The liquid container assembly according to claim 1, wherein the base assembly further comprises a thermoelectric conversion element thermally coupled to the container body and configured to generate electrical power based on a temperature difference between the container body and an environment surrounding the base assembly.

[Claim 12] A projection base module removably attachable to a container body for containing a liquid, the projection base module comprising:

a housing having a coupling portion configured to removably engage the container body;

a display module housed in the housing and configured to emit image light;

an optical member positioned above the display module in the housing and configured to deflect the image light emitted from the display module to form a virtual image upward;

a rechargeable battery and a wireless charging receiver housed in the housing;
and

wireless communication circuitry, a memory, and a controller housed in the housing, the controller being configured to receive image content from an external device via the wireless communication circuitry and to cause the display module to display the image content.

[Claim 13] The projection base module according to claim 12, further comprising a sealing member arranged to be compressed between the housing and the container body when the projection base module is coupled to the container body, thereby inhibiting ingress of liquid into the housing.

【書類名】 外国語要約書

[Problem] To provide a liquid container capable of presenting a floating (pseudo-holographic) image within the container while keeping the electronics separable from the liquid-contacting container body for washing, charging, and reuse.

[Solution] A liquid container assembly (100) includes a container body (1) and a base assembly (2) removably coupled thereto, with a sealing member (7) therebetween preventing liquid ingress. A display module (5) in the base assembly emits image light, and an optical member (6) redirects the image light so that a virtual image is perceived to float within or above the container body. The base assembly houses all electrically powered components, including a rechargeable battery (8), a wireless charging receiver (9), wireless communication circuitry (10), a controller (11), and a memory (12), such that the container body is free of electronics and displayed content can be updated wirelessly from an external device (200). The container body may take any form factor, such as a cup, glass, bottle, carafe, or pitcher.

[Selected Drawing] FIG. 1

FIG.1

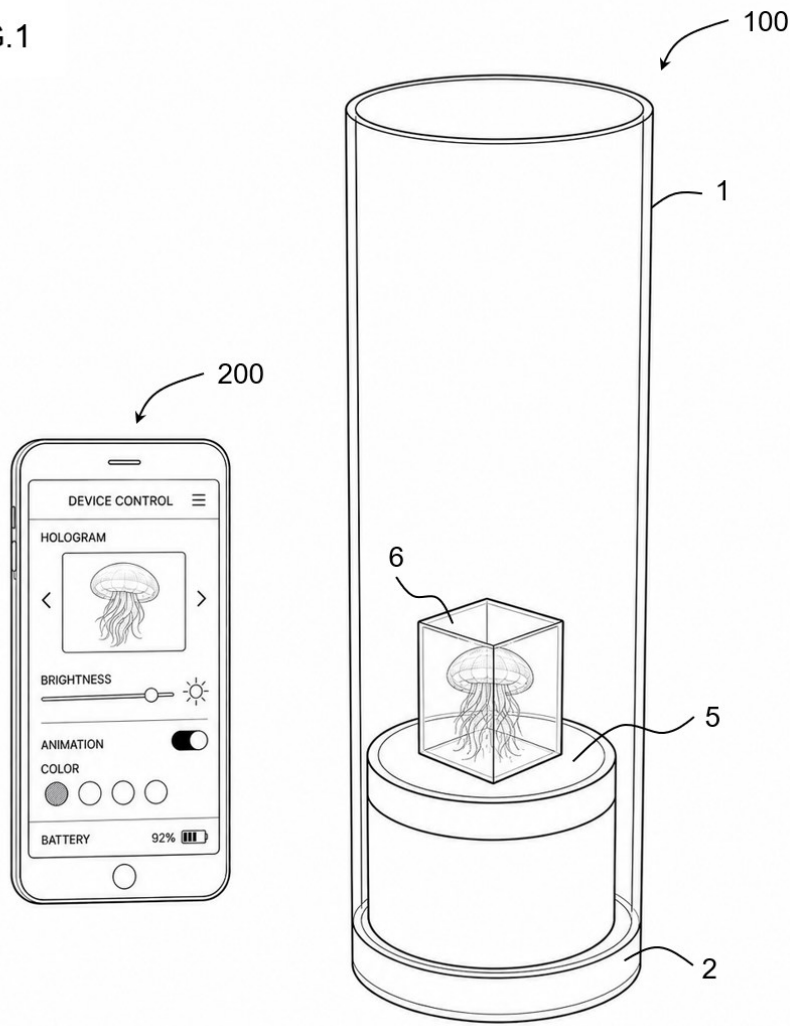


FIG.2

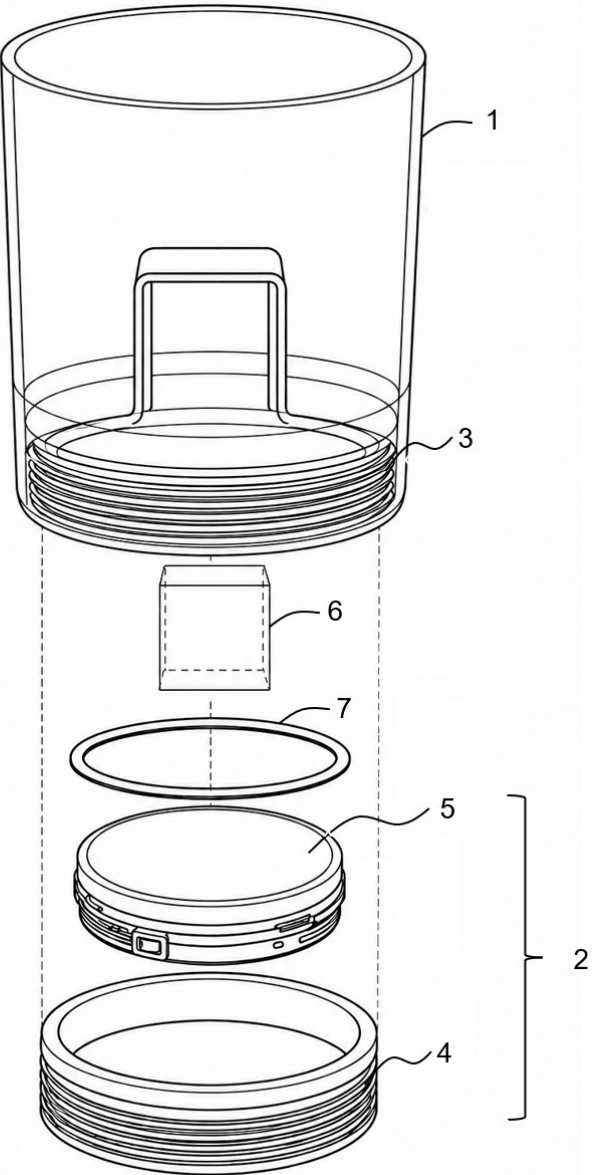


FIG.3

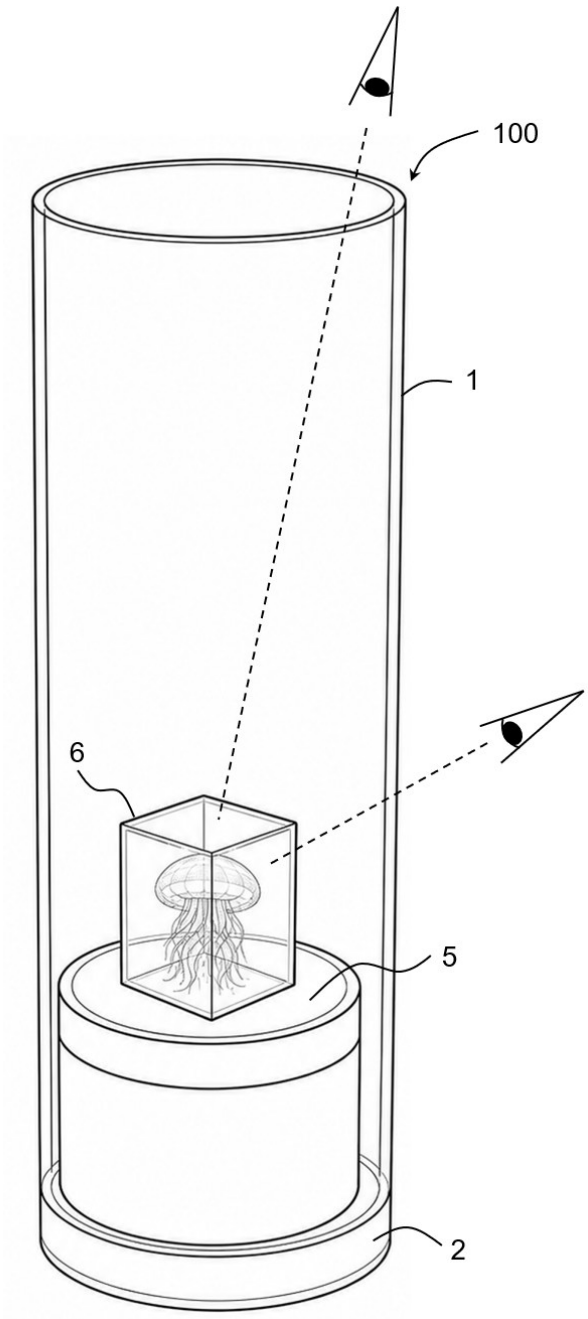


FIG.4

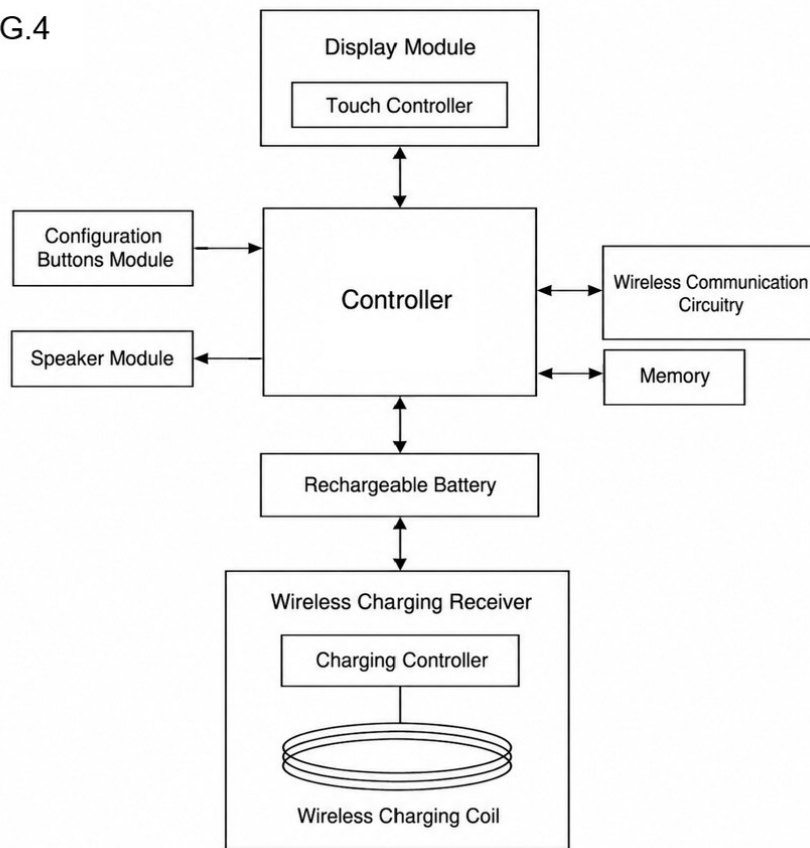


FIG.5

